

Table 1 Simulated example Cox hazard ratio (hr) estimates (denoted $\hat{\theta}$) for the ITT population and subgroups H and H^c . $\hat{\theta}(ITT)$ denotes the overall ITT population hr estimate and $\hat{\theta}(H)$ denotes the hr estimate based on the a-priori known (true) H subgroup $\hat{\theta}(\hat{H})$ denotes the hr estimate based on the estimated subgroup H and $\hat{\theta}^*(\hat{H})$ denotes the corresponding bootstrap bias-corrected estimate. Estimates for the complement H^c are defined analogously.

	HR Estimate	Lower	Upper	C(a-priori)	C(fixed)
ITT					
ITT	0.8376815	0.6873089	1.020953	.	.
H subgroup estimates					
H a-priori	2.35787	1.47058	3.780517	.	1
H estimate	3.077133	1.604064	5.902971	1	1
Bias-corrected H estimate	2.233123	1.456923	3.422858	1	1
H-complement subgroup estimates					
H^c a-priori	0.6721508	0.5372828	0.8408731	.	1
H^c estimate	0.7330196	0.5924729	0.9069068	1	1
Bias-corrected H^c estimate	0.7438026	0.5929508	0.9330324	1	1

As an illustration we consider a simulated dataset generated as described in Section 3 below. In this simulated example there are $n = 700$ subjects, randomized 1:1, where the underlying (marginal) hazard ratio for the harm population H is 2.5, and for the complement H^c , the (marginal) hazard ratio is 0.65. The overall (marginal) hazard ratio is 0.71. Here the actual number of subjects in the H subgroup is 101 and the number of subjects in the complement equals 599.

Applying our novel procedure the estimated subgroup $\hat{H}$ consists of 64 subjects and thus 636 in the estimated complement $\hat{H}^c$. The number of subjects in H that are included in $\hat{H}$ is 52 (i.e., $|H \cap \hat{H}| = 52$): Of the 101 subjects in H , $52/101 = 51\%$ are correctly classified and of the 64 subjects in $\hat{H}$, $52/64 = 81\%$ are correctly classified. For the complement, the number of subjects in H^c that are included in $\hat{H}^c$ is 587: Of the 599 subjects in H^c , $587/599 = 98\%$ are correctly classified and of the 636 subjects in $\hat{H}^c$, $587/636 = 92\%$ are correctly classified.

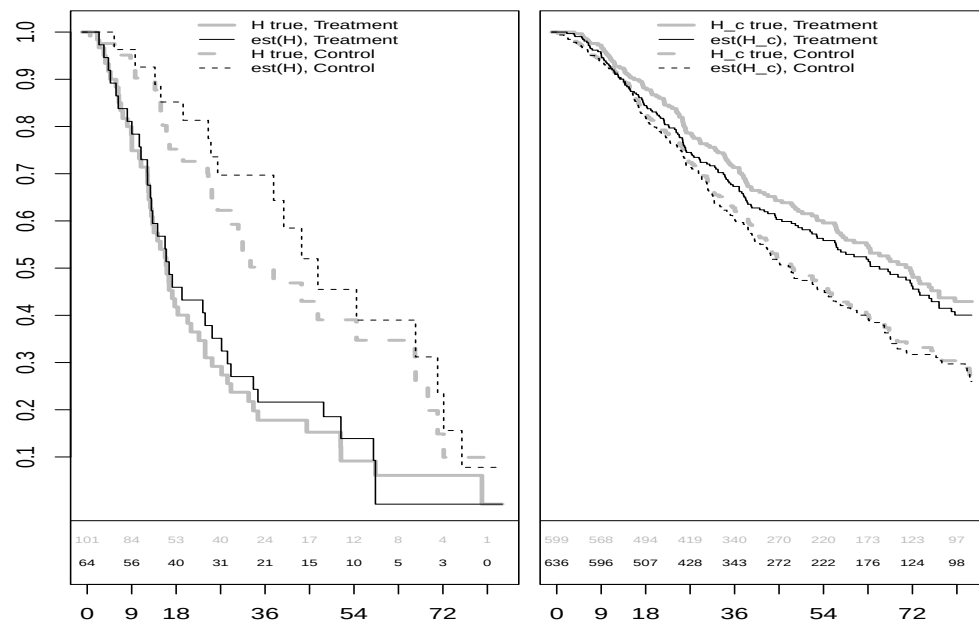


Fig. 1 Simulated example